

*the adverse impact of volatility. The Kelly criterion defines this threshold. The Kelly criterion indicates that the fraction that should be wagered to maximize compounded return over the long run equals:*

$$F = P_w - (P_L/W)$$

*where*

*F = Kelly criterion fraction of capital to bet*

*W = Dollars won per dollar wagered (i.e., win size divided by loss size)*

*P<sub>w</sub> = Probability of winning*

*P<sub>L</sub> = Probability of losing*

*When win size and loss size are equal, the formula reduces to:*

$$F = P_w - P_L$$

*For example, if a trader loses \$1,000 on losing trades and gains \$1,000 on winning trades, and 60 percent of all trades are winning trades, the Kelly criterion indicates an optimal trade size equal to 20 percent ( $0.60 - 0.40 = 0.20$ ).*

*As another example, if a trader wins \$2,000 on winning trades and loses \$1,000 on losing trades, and the probability of winning and losing are both equal to 50 percent, the Kelly criterion indicates an optimal trade size equal to 25 percent of capital:  $0.50 - (0.50/2) = 0.25$ .*

*Proportional overbetting is more harmful than underbetting. For example, betting half the Kelly criterion will reduce compounded return by 25 percent, while betting double the Kelly criterion will eliminate 100 percent of the gain. Betting more than double the Kelly criterion will result in an expected negative compounded return, regardless of the edge on any individual bet.*

*The Kelly criterion implicitly assumes that there is no minimum bet size. This assumption prevents the possibility of total loss. If there is a minimum trade size, as is the case in most practical investment and trading situations, then ruin is possible if the amount falls below the minimum possible bet size.*

*I learned about the Kelly criterion from Claude Shannon back at MIT. Shannon had worked with Kelly at Bell Labs. I guess Shannon was the leading light at Bell Labs and Kelly was perhaps the second most*